Anomaly Detection for Performance Rewards and Correction in Mixed Reality Collaboration

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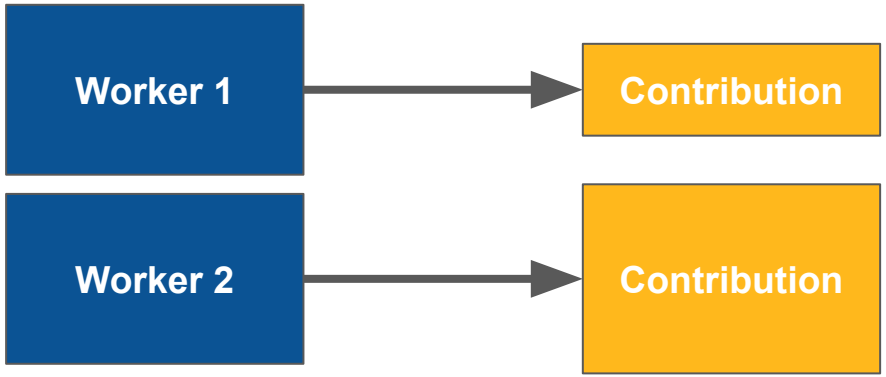
Motivation and Problem Definition

Motivation

- Efficiently evaluating group collaboration in Mixed Reality can improve company performance and outcomes.
- By analyzing key interaction dynamics, attention, communication, spatial coordination, and engagement, organizations can provide corrective feedback or reward high-performing teams.



Problem Definition

Objective: Detect deviations from normal collaborative behavior in MR to guide corrective action or reward allocation

Challenges:

1. **Complex, high-dimensional signals** → gaze, motion, communication

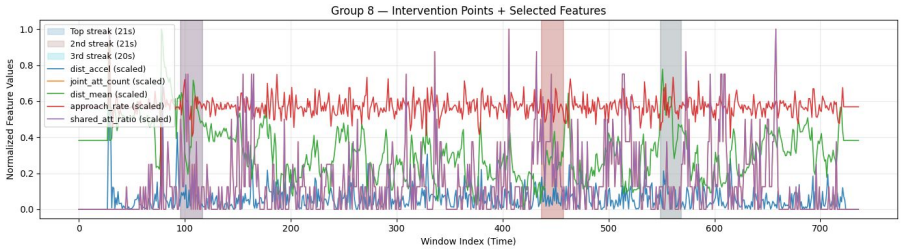
2. **Limited datasets** → supervised methods prone to overfitting

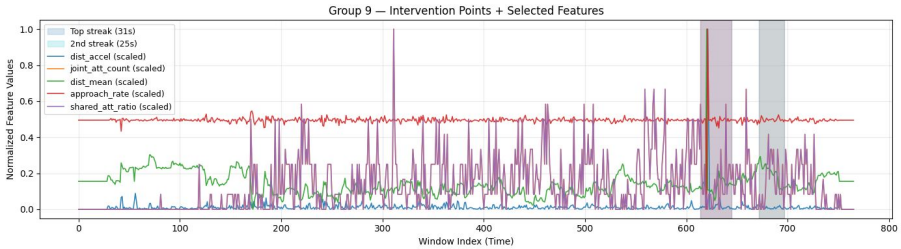
3. **Emergent group-level faults** →communication failures, spatial disengagement

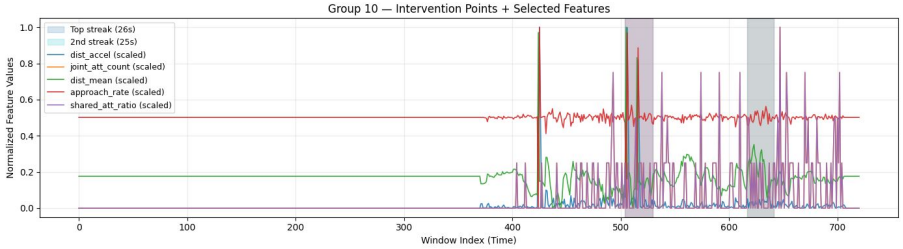
Technical Approach

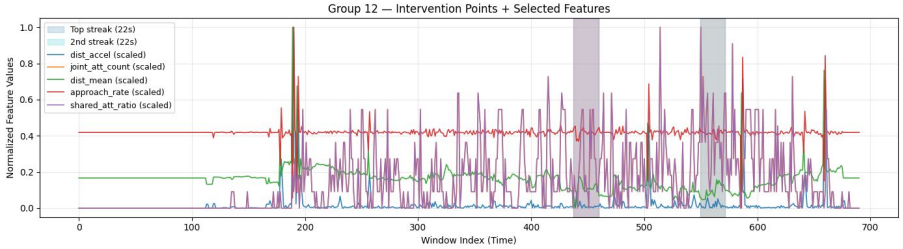
How do we find optimal intervention points?

- **Anomaly Detection On Pairwise Outputs**
  - Per second dyadic metrics, measuring interactions between pairs of teammates
  - Anomalous time windows = potential points of correction









Methods of Anomaly Detection

- **Density-/Proximity- based methods (local outlier, connectivity-based outlier, local correlation integral, histogram-based outlier score)**
  - Performs well on smaller datasets, fast
  - Distance / density estimation is expensive in high dimensions
- **Deep-learning methods (LSTM Autoencoder, DeepAnT, FuseAD)**
  - Best for modeling time dependent behavior (M. Munir, M. A. Chattha, A. Dengel and S. Ahmed)
  - Most accurate
  - Needs large dataset, slow
- **Isolation Forest**
  - Performs well on smaller datasets, fast
  - Well suited for finding strong outliers in multidimensional datasets
  - Can produce different results across runs unless seeded

Methods of Anomaly Detection

Liu, F. T., Ting, K. M., and Zhou, Z.-H. 2012:

Most existing anomaly detection approaches, including classification-based methods [Abe et al. 2006; Shi and Horvath 2006], Replicator Neural Network (RNN) [Williams et al. 2002], one-class SVM [Tax and Duin 2004] and clustering-based methods [He et al. 2003], construct a profile of normal instances, then identify anomalies as those that do not conform to the normal profile. Their anomaly detection abilities are usually a 'side-effect' or by-product of an algorithm originally designed for a purpose other than anomaly detection (such as classification or clustering). This leads to two major drawbacks: (i) these approaches are not optimized to detect anomalies—as a consequence, these approaches often under-perform resulting in too many false alarms (having normal instances identified as anomalies) or too few anomalies being detected; (ii) many existing methods are constrained to low dimensional data and small data size because of the legacy of their original algorithms.

Methods of Anomaly Detection

Liu, F. T., Ting, K. M., and Zhou, Z.-H. 2012:

Apart from the key difference in using isolation as the means to detect anomalies, *i*Forest is distinguished from existing model-based (e.g. [Abe et al. 2006; He et al. 2003]), link-based (e.g. [Ghoting et al. 2004]), depth-based (e.g. [Rousseeuw and Leroy 1987]), distance-based (e.g. [Knorr and Ng 1998]) and density-based methods (e.g. [Breunig et al. 2000]) in the follow ways:

I The characteristic of isolation trees enables them to exploit subsampling to an extent that is not feasible in existing methods (more details are provided in Section 5.5).

II *i*Forest utilizes no distance or density measures to detect anomalies. This eliminates a major computational cost of distance calculation in all distance-based and density-based methods.

III *i*Forest has a linear time complexity with a small constant and a minimal memory requirement; it is an algorithm with constant training time and space complexities. To the best of our knowledge, the best performing existing method achieves only approximate linear time and space complexities [Angiulli and Fassetti 2009].

IV *i*Forest has the capacity to scale up to handle extremely large data size and high-dimensional problems with a large number of irrelevant attributes (Section 5.6).

- Best method for our application: **Isolation Forest**

Isolation Forest

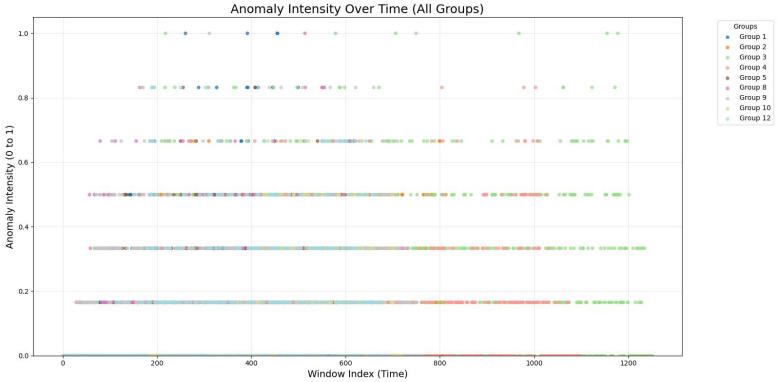
- **Unsupervised anomaly detection algorithm**
- **Focuses on isolating anomalies rather than profiling normal data**
  - Assumes anomalies will significantly deviate from other data points
- **Creates a forest of isolation trees**
  - Isolation trees isolate individual data points by repeatedly splitting data
    - Data is split based on randomly chosen features and split values
      - At each node, a random feature from the dataset is selected
      - For that feature, a random split value within range of the feature's values is chosen
        - Data point is sent to left or right branch based on comparison to split value
        - Process continues recursively until data point reaches a leaf node
          - Leaf node → isolated data point
    - Randomness ensures that anomalies do not get hidden within specific branches of the tree

Isolation Forest Results

- **Anomaly Score**
  - Based on length of path required to isolate datapoint
  - Anomalies should be easier to isolate → shorter path
  - Normal data points → longer path
  - Anomaly score = average path length across all isolation trees in the forest for a specific data point
- **Anomaly Label**
  - -1 = Anomaly
  - 1 = Normal

How We Use This Data

- **Anomaly Intensity** - [0% to 100%] Average of the Anomaly Labels of the Whole Group at that Second
- **Intervention Points**
  - Largest Unbroken Streak of Anomaly Intensity over 50% (gaps of 5 seconds allowed)
  - Any Anomaly Intensity over 50% over 20 seconds

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===== Longest Streaks with Threshold & Allowed Gaps =====

Group 1: Longest streak = 30 sec (from window 318.0 to 347.0) (threshold=0.5, max_gap=5)
Group 2: Longest streak = 19 sec (from window 272.0 to 290.0) (threshold=0.5, max_gap=5)
Group 3: Longest streak = 43 sec (from window 330.0 to 372.0) (threshold=0.5, max_gap=5)
Group 4: Longest streak = 46 sec (from window 971.0 to 1016.0) (threshold=0.5, max_gap=5)
Group 5: Longest streak = 15 sec (from window 508.0 to 522.0) (threshold=0.5, max_gap=5)
Group 8: Longest streak = 41 sec (from window 430.0 to 470.0) (threshold=0.5, max_gap=5)
Group 9: Longest streak = 25 sec (from window 620.0 to 644.0) (threshold=0.5, max_gap=5)
Group 10: Longest streak = 28 sec (from window 503.0 to 530.0) (threshold=0.5, max_gap=5)
Group 12: Longest streak = 22 sec (from window 550.0 to 571.0) (threshold=0.5, max_gap=5)

===== Sustained (20s+) Anomaly Streaks per Group =====

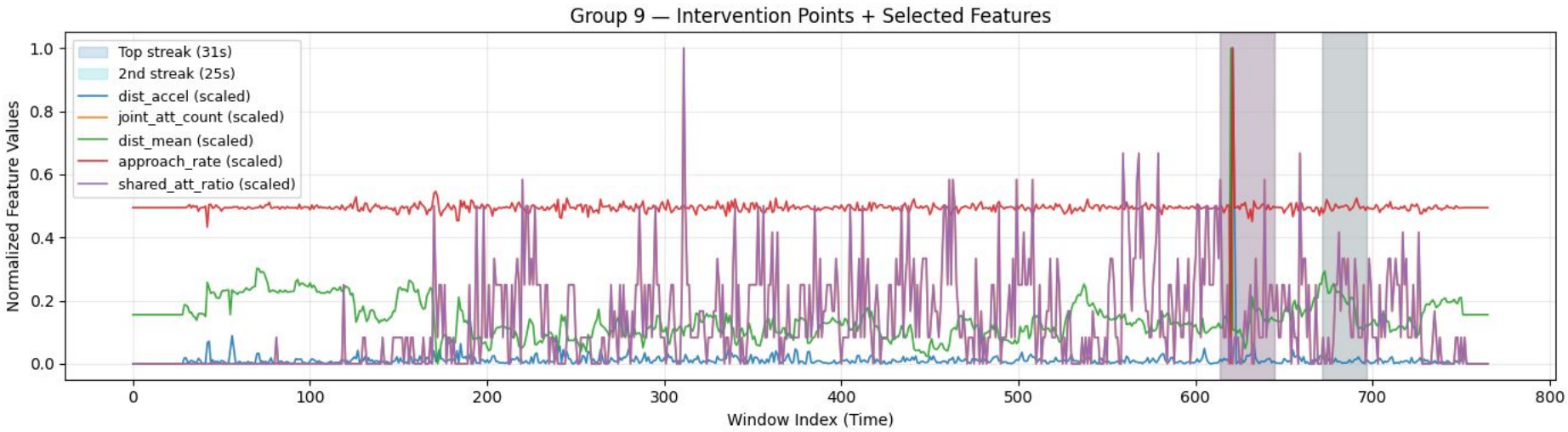

Group 1: Found 4 streak(s) >= 20 seconds
  - Streak of 30 sec from window 318.0 to 347.0
  - Streak of 29 sec from window 376.0 to 404.0
  - Streak of 22 sec from window 439.0 to 460.0
  - Streak of 20 sec from window 488.0 to 507.0
Group 2: No streaks >= 20 seconds.

Group 3: Found 11 streak(s) >= 20 seconds
  - Streak of 43 sec from window 330.0 to 372.0
  - Streak of 40 sec from window 437.0 to 476.0
  - Streak of 29 sec from window 493.0 to 521.0
  - Streak of 33 sec from window 561.0 to 593.0
  - Streak of 21 sec from window 613.0 to 633.0
  - Streak of 30 sec from window 646.0 to 675.0
  - Streak of 39 sec from window 678.0 to 716.0
  - Streak of 25 sec from window 931.0 to 955.0
  - Streak of 21 sec from window 966.0 to 986.0
  - Streak of 21 sec from window 1077.0 to 1097.0
  - Streak of 21 sec from window 1168.0 to 1188.0

Group 4: Found 1 streak(s) >= 20 seconds
  - Streak of 46 sec from window 971.0 to 1016.0
Group 5: No streaks >= 20 seconds.

Group 8: Found 2 streak(s) >= 20 seconds
  - Streak of 41 sec from window 430.0 to 470.0
  - Streak of 24 sec from window 546.0 to 569.0
```

Example Output

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Group 9: Longest streak = 31 sec (from window 614.0 to 644.0)
```
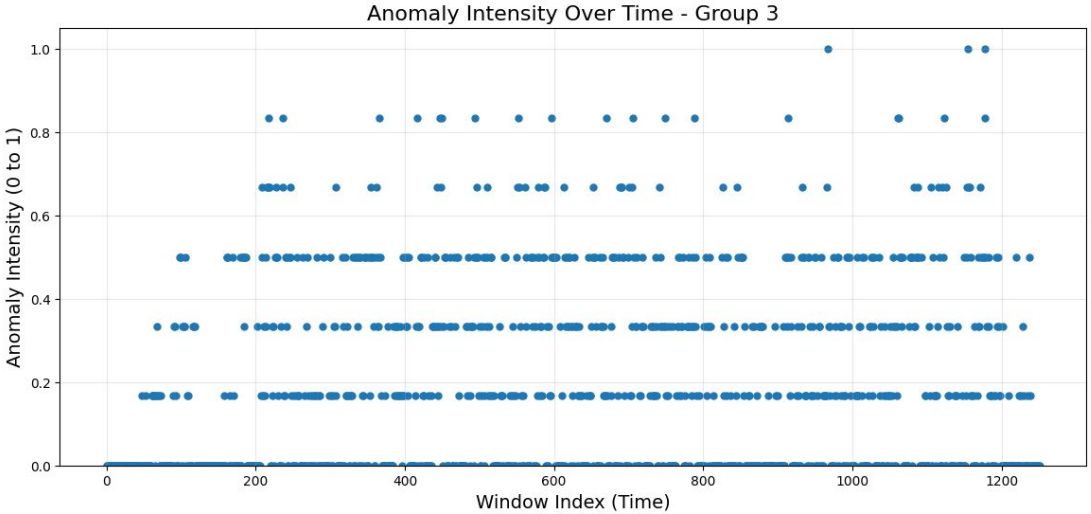
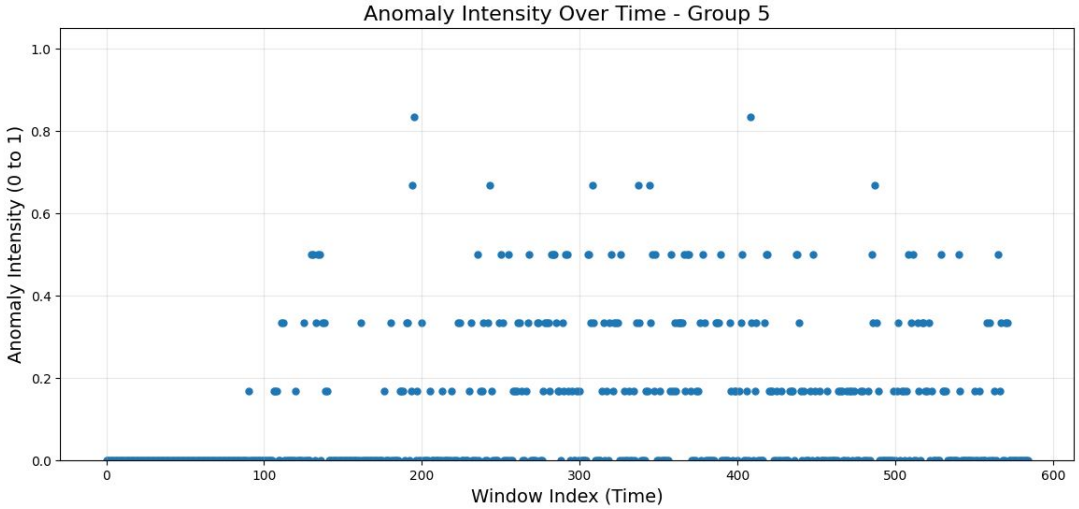

```
Group 9: Found 2 streak(s) >= 20 seconds
  - Streak of 31 sec from window 614.0 to 644.0
  - Streak of 25 sec from window 672.0 to 696.0
```





Group Comparison

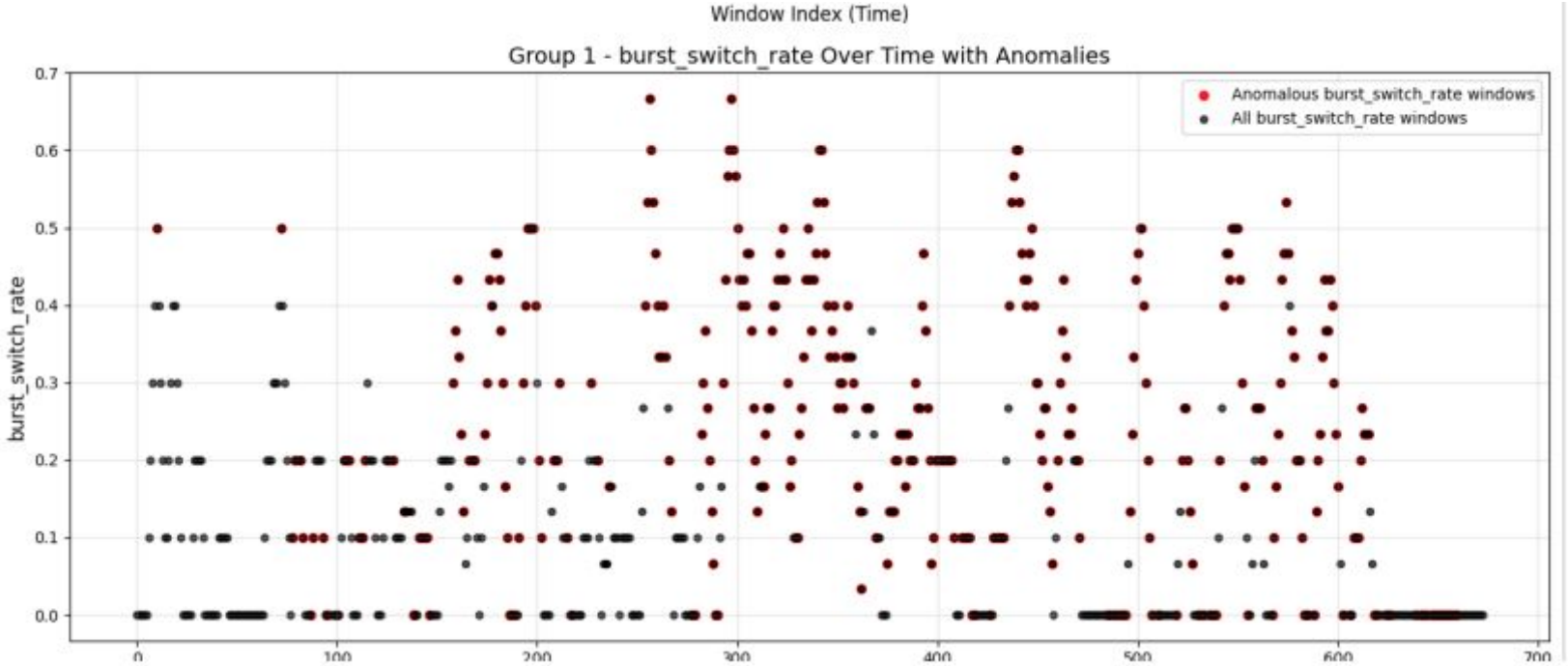
- Group 3 Takes 2x Time

```
Group 3: Found 8 streak(s) >= 20 seconds
  - Streak of 43 sec from window 330.0 to 372.0
  - Streak of 37 sec from window 440.0 to 476.0
  - Streak of 33 sec from window 561.0 to 593.0
  - Streak of 20 sec from window 613.0 to 632.0
  - Streak of 30 sec from window 646.0 to 675.0
  - Streak of 39 sec from window 678.0 to 716.0
  - Streak of 21 sec from window 1077.0 to 1097.0
  - Streak of 21 sec from window 1168.0 to 1188.0

Group 4: Found 1 streak(s) >= 20 seconds
  - Streak of 46 sec from window 971.0 to 1016.0
Group 5: No streaks >= 20 seconds.
```

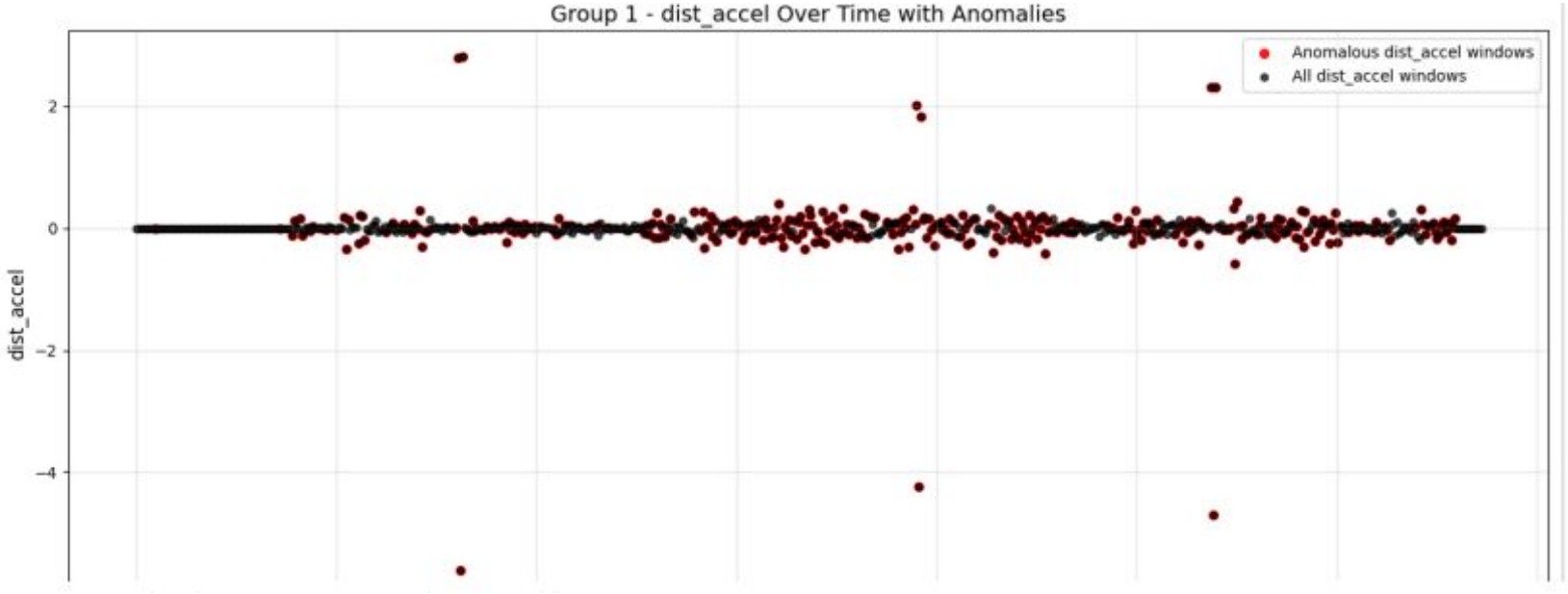
What data do we use?

- **Pairwise data contains per second measurements of behavior**
- **Literature supports use of features in specific categories**
  - **Spatial Interaction**
    - Proxemics: study of how people use space and its effects on behavior and social interaction
    - Theory of proxemics has been used for designing VR experiences (Kim, I., & Sung, J. (2024)).
  - **Shared Attention**
    - Studies have been conducted to explore relationship between communication and visual attention
    - Measure of shared visual attention shown to reflect success of communication (Richardson, D. C., & Dale, R. (2005))
- **Exclude data that is not ideal for isolation forest**

Un-Ideal


Window Index (Time)
Group 1 - burst_switch_rate Over Time with Anomalies
burst_switch_rate
Anomalous burst_switch_rate windows
All burst_switch_rate windows
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
100
200
300
400
500
600
700

Ideal

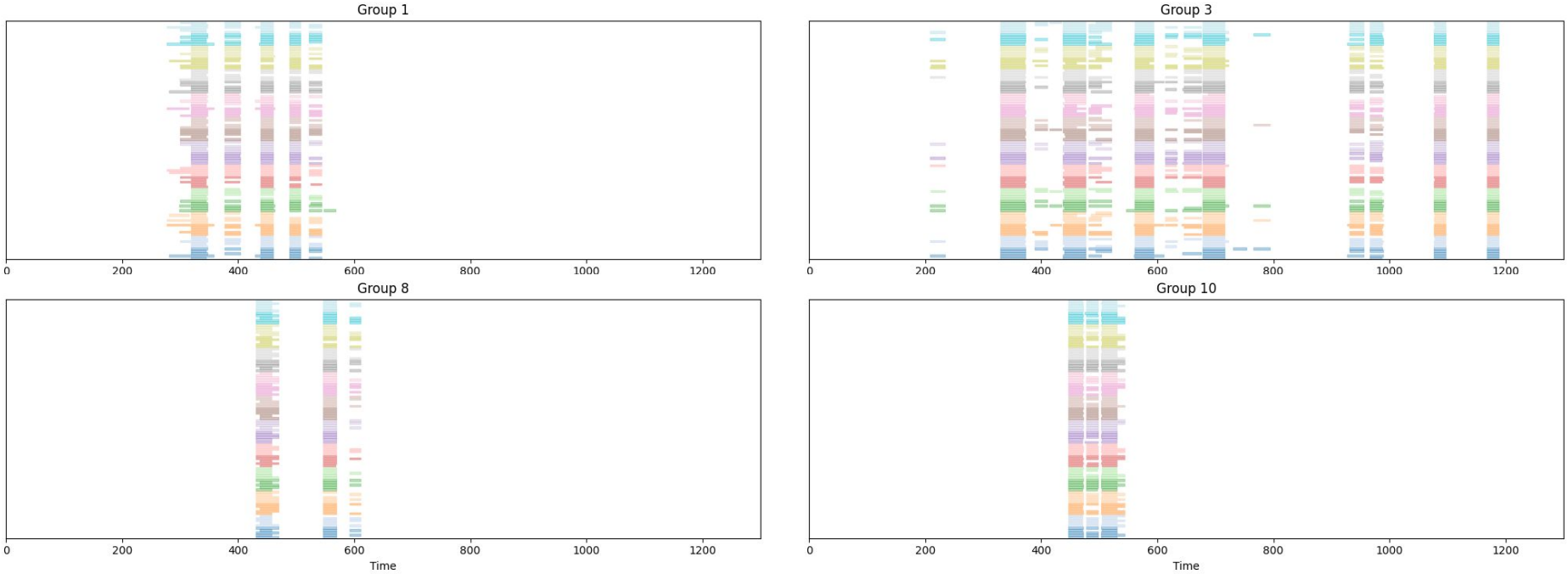


- **Chosen features:**
  - Dist_accel, Dist_mean, Approach_rate, Shared_att_ratio, Joint_att_count

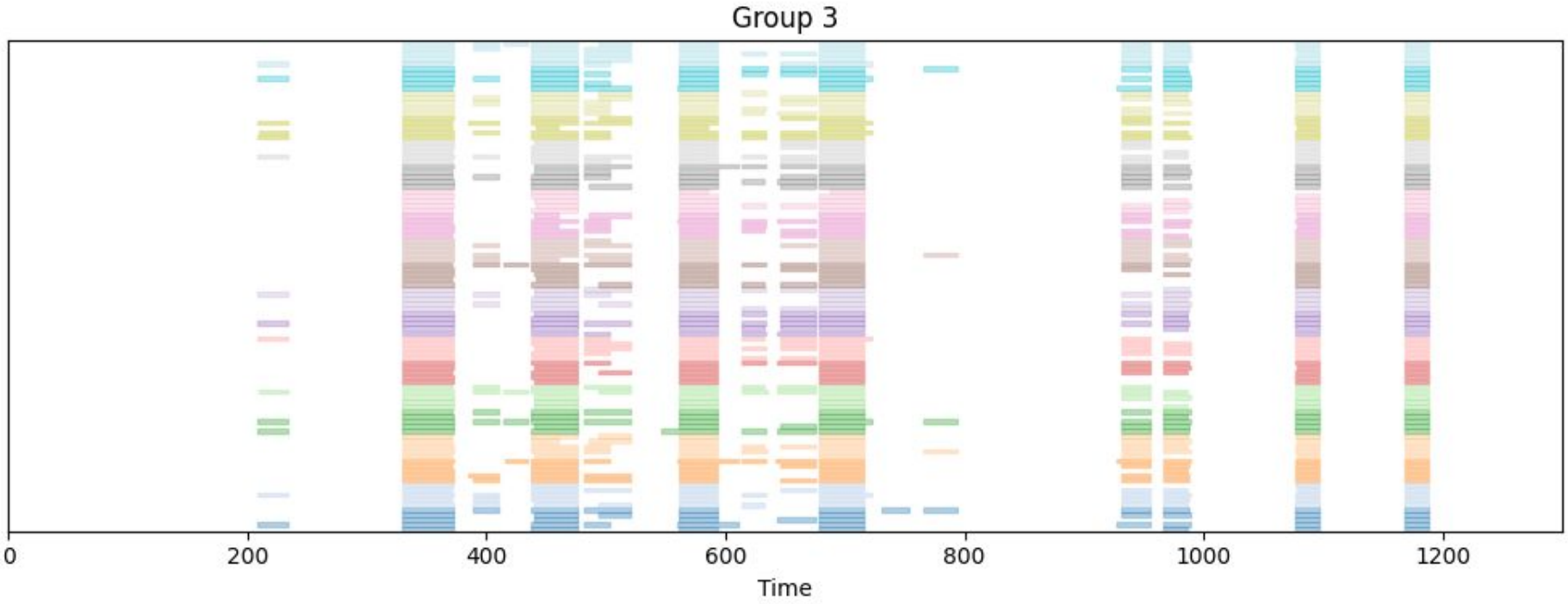
Evaluation

Score Stability

- Isolation Forests are inherently based off randomness, so running our algorithm repeatedly (100x) with different random seeds validated our data



Score Stability

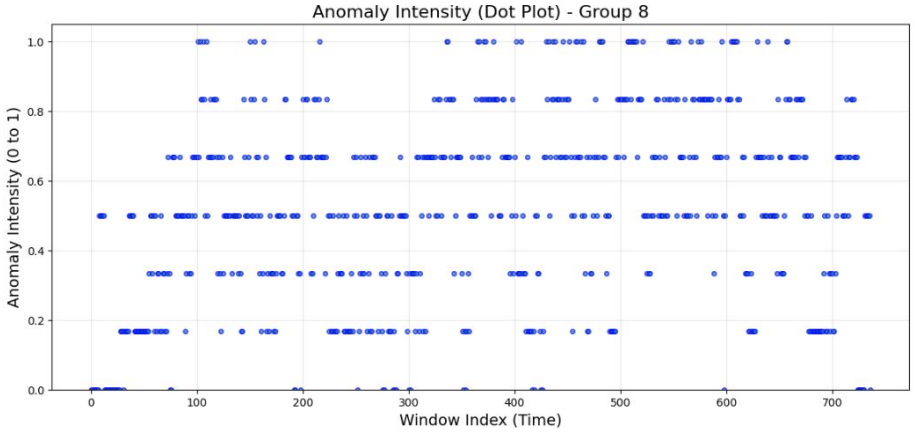


Data Integrity Check

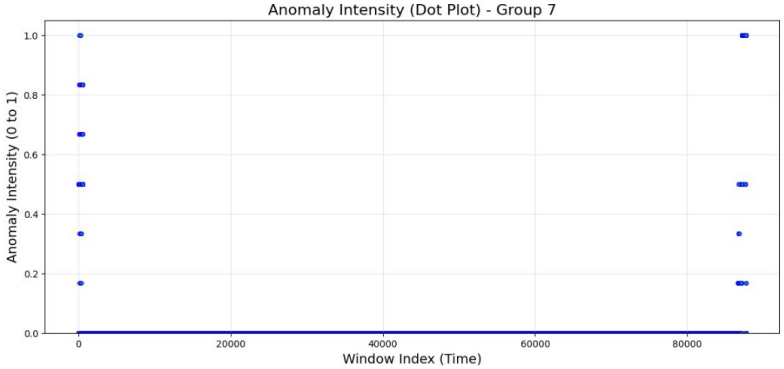
Data Fed to Algorithm May be faulty

- Group 7 and 11 went for more than 22 hours
- Group 6 has long period of inactivity

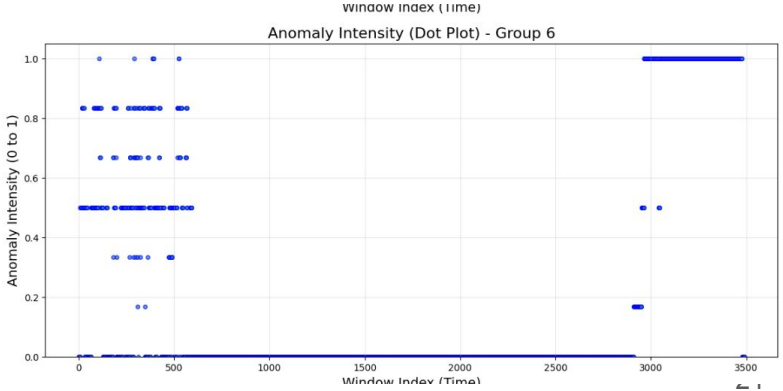
Normal Data Examples



Bad Data Examples



Long Period of Inactivity



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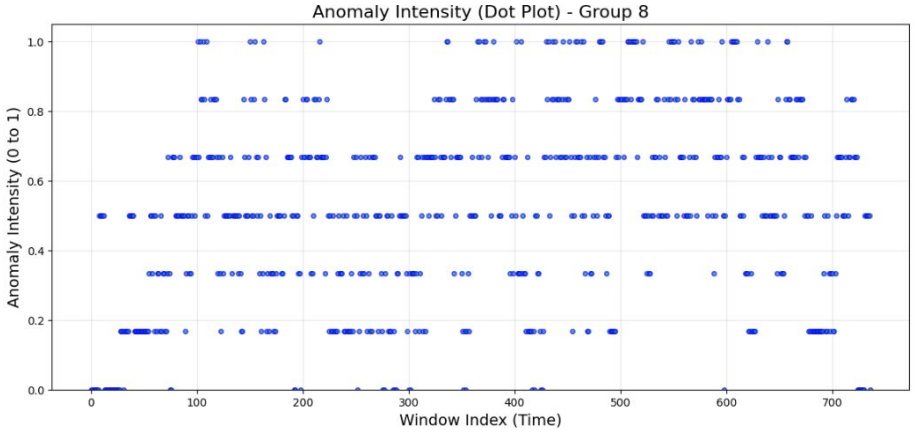

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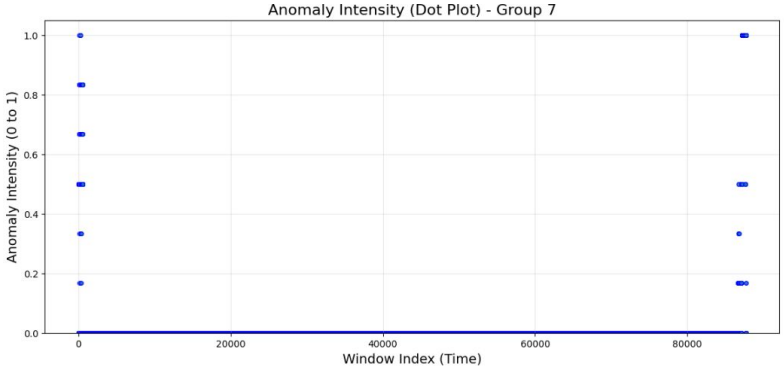

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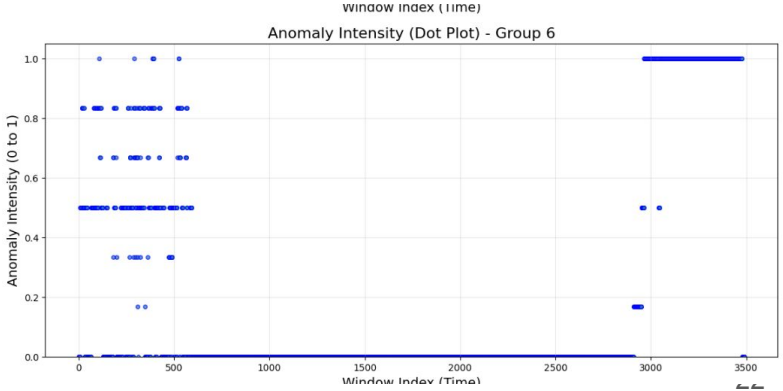
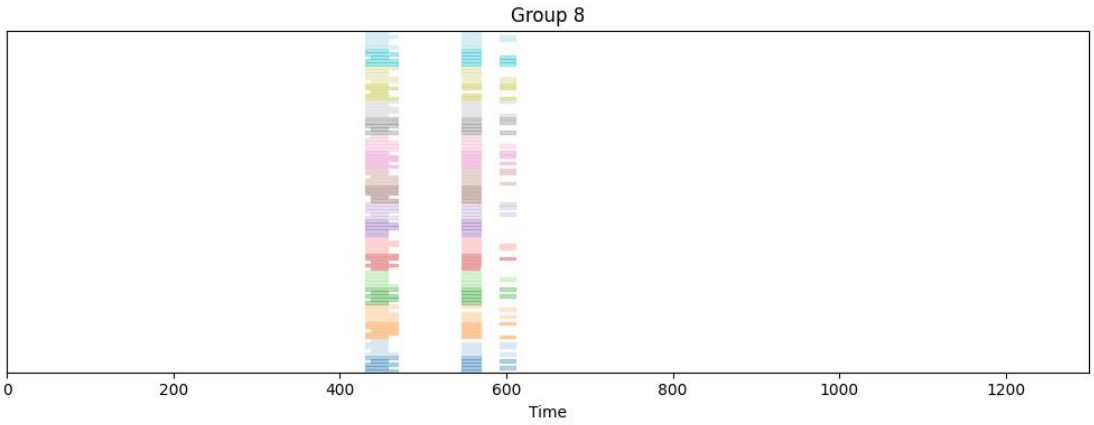


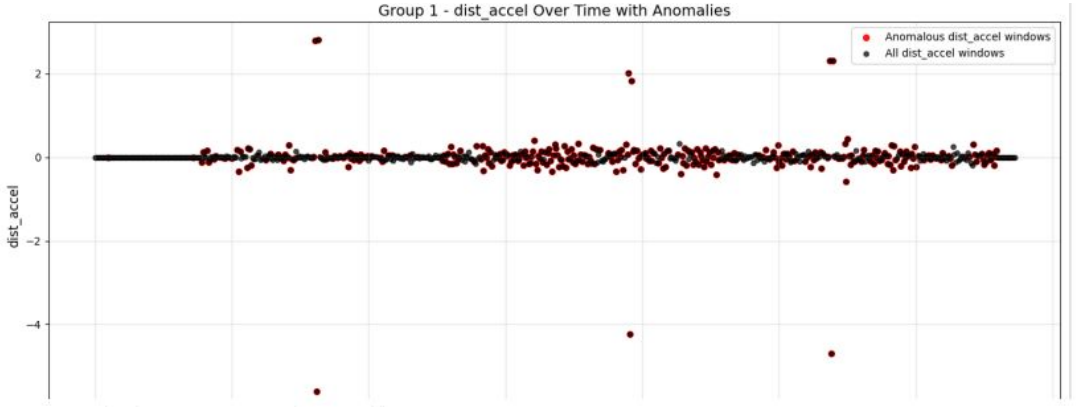
Discussion of the Results

Results and Comparison

Detection + Validation

- **High-fidelity intervention windows identified:**
- **Group-level aggregation improves interpretability:**
- **Sustained disengagement captured (not noise):**
- **Strong visual + algorithmic validation:**





Data Auditing + Model Reliability

- **Critical role of Data Auditing:**
- **Model reliability improved after exclusion:**
- **Data quality > Model tuning**
- **Isolation Forest validated for MR timing tasks:**
  - Despite being non-temporal, Isolation Forest supported reliable **sequential intervention timing** when paired with windowed scoring, normalization, and group aggregation.
- **Best suited for early-stage MR analytics:**
  - Effective for **small datasets** where **intervention timing** is more critical than generative sequence modeling.

Key Takeaways

- Simple models can still support **reliable, time-sensitive intervention** when paired with good windowing and aggregation
- **Data quality matters more than model tuning** — bad distributions break learning
- **Group-level anomaly signals** are more meaningful than individual labels
- **Visual + algorithmic validation together** build confidence in the results
- Isolation Forest is **well-suited for early-stage, small-dataset MR analytics**

Future Work

Future Work

1. **Larger and More Diverse Datasets** – Future deployments should incorporate multiple session MR datasets to validate the model across multiple groups and environments.

2. **Task-Aware Hybrid Detection** – Although the current system is unsupervised, incorporating supervision could reduce false positives and distinguish between *productive silence* and *true disengagement*.

3. **Real-Time Feedback Interface** – Integrating the anomaly intensity signal into MR dashboards, adaptive instructors, or AI agents enables live interventions such as attention re-anchoring cues, spatial guidance overlays, or automated encouragement systems.

References

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Thank You!